

Co-Design of Detection, Actuation, and Control for Solid-State Circuit Breakers in Electrified Aircraft Propulsion System

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Abstract—As a crucial component of the direct current (dc) electrified aircraft propulsion (EAP) systems, the dc solid-state circuit breaker (dc-SSCB) has been extensively studied at the breaker level, including topology, energy absorption circuit, gate driver, and control. However, system-level considerations are also crucial but have received limited attention. This article presents a comprehensive solution with integrated circuitries functioning as fault detectors, actuators for regulating semiconductor operating status, and coordinated control to seamlessly transition between modes (e.g., the normal mode, precaution mode, i²t mode, and instantaneous trip mode). This design enables the developed SSCB to limit system fault currents within the short-circuit current rating (SCCR) and respond appropriately to various fault scenarios. First, two improved fault detection circuits are introduced to enhance SSCB applications. Second, four operation modes are proposed to achieve optimal system functionality. Third, leveraging these four operation modes, a control strategy is demonstrated to execute different actions based on the fault scenarios. Finally, the feasibility and effectiveness of the proposed comprehensive SSCB are validated through experimental results.

Index Terms—Solid-state circuit breaker, electrified aircraft propulsion (EAP), current limitation, direct current (dc).

I. INTRODUCTION

WITH the advancement of power electronics, the direct current (dc) distribution system has shown significant potential to replace the alternating current (ac) distribution system in electrified aircraft propulsion (EAP) systems [4], [5].

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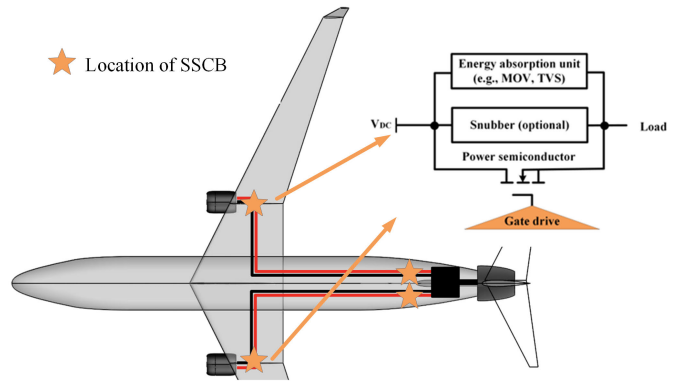


Fig. 1. Basic topology and location of SSCB in EAP systems [16].

Compared to ac distribution systems, dc distribution systems are more compatible with battery energy storage systems (BESSs), which is essential for EAP systems. Meanwhile, the dc distribution system offers higher efficiency, which can help EAP systems reduce overall weight. DC cables can carry more power than ac cables with the same root-mean-square (rms) current. In addition, the absence of the skin effect and reactive power flow results in lower power loss in dc cables.

From another perspective, the developing dc EAP system with a high penetration of power electronics imposes more stringent requirements for fault-tolerant systems [8], [9]. For instance, power electronics systems are highly susceptible to high current values, making it crucial to have a dc circuit breaker (CB) with a fast response time to interrupt overcurrent [11]. In addition, due to the fragility of power converters, dc-CBs must also be capable of lowering the system's fault current. The fault ride-through capability of dc-CBs is also essential for ensuring reliable system operation. However, traditional protection systems are not designed to accommodate power systems with a high penetration of power electronics, highlighting the need for new protection mechanisms to serve future EAP systems better.

To overcome the challenges of the dc distribution system for EAP systems, dc-SSCBs have become a significant research focus [14], [15]. Unlike mechanical and hybrid CBs, dc solid-state CBs (dc-SSCBs) rely on power semiconductors to conduct normal system current and quickly interrupt fault currents, offering significantly faster response times. Fig. 1

TABLE I
COMPARISON WITH OTHER SSCB

	Rating	Interrupti on capability	Number of operation mode	Current limitation capability	Response speed	Fault ride-through capability
Ours	400V/15A GaN	150A +	4	Actively controlled by gate voltage	~50ns for PreDeSat	Yes
[1]	380V/20A GaN	70A	3	Yes	us level	No
[2]	200V/12A Thyristor	120A	1	No	us level	No
[3]	400V/14A IGBT + SCR	20A	1	No	us level	No
[6]	180V/3A GaN	20A	1	No	282 ns	No
[7]	200V/15A MOSFET	30A	1	No	us level	No

displays a representative topology of the dc-SSCB and their locations in the EAP system of Single-Aisle Turboelectric Aircraft with Aft Boundary Layer Propulsion (STAR-ABL) [16]. Some designs position the SSCB on both rails. The advantage of installing an SSCB on each rail is that it allows the system to handle single-point ground faults [4], [5]. As a flight-critical component in the EAP system, the MVdc system must continue operating at full power, even with a ground fault. This requires the SSCB to clear the fault current if a second ground fault occurs. However, placing SSCBs on both dc rails doubles the power loss and weight at each bus location, making it more challenging to achieve efficiency and specific power targets.

A dc-SSCB includes a power semiconductor, a snubber circuit, and an energy absorption unit. Due to their low conduction loss and high switching speed, wide bandgap power semiconductors are ideal candidates for SSCBs [17].

Since NASA invented its first dc-SSCB in 1973 (based on the online library) [18], the study on SSCB mainly focuses on power semiconductor selection, topology, energy absorption circuit and control design, fault detection design, etc. Meanwhile, as an active participant in the distribution system's reliable operation, beyond the primary fault interruption duty, SSCB should be capable of performing other system-friendly functions that benefit the system's operation. Moreover, Table I shows the comparison among some SSCB for modular SSCB. Most of them focus on the basic interruption function of the SSCB.

A few studies have explored enhanced functions for dc-SSCBs, including soft start-up [19], fault location detection [20], fault current limiting [21], [22], and soft reclosing [23]. Take current limiting as an example, enabling dc-SSCBs with inrush current limitations during system start-up is preferred. One feasible method is to decrease the gate voltage once an overcurrent event is detected [21], [22]. The fault current can force the power semiconductor to enter the saturation region, and the $V-I$ curve determines the system's current limitation level. To improve the current limiting capability of the semiconductor in the saturation region, a current limiting strategy for SSCB with series-connected switching cells [24]. However, these studies primarily focus on the current limitation capacity of the SSCB, lacking a holistic control strategy to coordinate the limiter and breaker capabilities for optimal performance in an EAP protection system.

It is also essential for dc-SSCBs to distinguish between overcurrent and SC conditions. He et al. [25] propose a digitally controlled current-time profile-based dc-SSCB for both overcurrent and SC protection. The proposed current-time profile helps the dc-SSCB avoid faults triggered by inrush currents during the start-up of power-electronics-intensive systems. However, this approach requires an additional micro-controller to control the SSCB, which introduces complexity and additional costs.

To enable reliable operation, the EAP system with high penetration of power electronics is more demanding for the SSCB-based protection system.

- 1) The fault current cannot exceed short-circuit current rating (SCCR) defined by the system (e.g., $10\times$) [26].
- 2) Limit the worst scenario fault current below SCCR and trip the SSCB instantaneously (e.g., sub- μ s).
- 3) Interrupt the relatively low current fault, instead of instantaneous response, following the i^2t capability of SSCB for better fault type self-awareness and system protection coordination.
- 4) Stand by and bypass a temporary fault with self-recovery capability.

To satisfy these requirements, a system-friendly SSCB enabled by codesigning detection, actuation, and control is proposed in this article. The proposed system-friendly SSCB aims to provide an all-in-one solution with: a) integrated circuitries to serve as fault detectors (not only to detect the fault but also predict the fault current considering absolute fault current and di/dt) and actuator (to manage power device's operating status); and b) coordinated control to seamlessly transition from one mode to another (normal mode, precaution mode, i^2t mode, and instantaneous trip mode). For the fault detector, by introducing the influence of Ldi/dt , an improved fault detection circuit is proposed, which considers the value and the increasing ratio of the system's current. Table II also compares the traditional fault detection methods for SSCB. Meanwhile, for the actuator, the precaution mode can ensure the system's current cannot be higher than SCCR; i^2t mode guides SSCB's response to the low current fault based on the power device's thermal characteristics [27]; instantaneous trip mode is leveraged to interrupt the system's fault current. For the coordinated control, based on the different fault system scenarios, the proposed hardware-based controller can guide four different modes to enable SSCB to response accordingly.

TABLE II
COMPARISON OF DIFFERENT FAULT DETECTION METHODS FOR SSCB

Sensing technology	Advantages	Disadvantages
Current sensor [10]	✓ Fixed trigger current ✓ No extra loss	• Temperature sensitive • Limited bandwidth • Expensive
Shunt resistor [12]	✓ Fast response	• Voltage drop extra loss on the resistor • No galvanic isolation
Desaturation detection [13]	✓ Fast response ✓ No extra loss	• Trade-off between noise immunity and response time

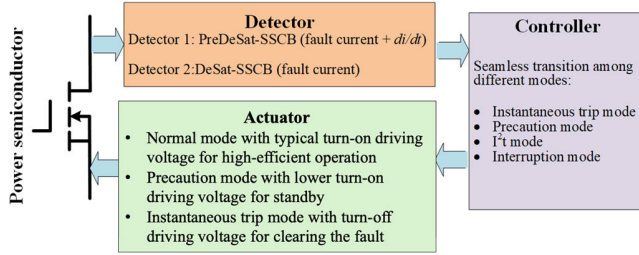


Fig. 2. System configuration of the proposed SSCB.

In summary, the contributions of this article include the follows.

- 1) A novel control strategy is introduced to coordinate four distinct operational modes for the SSCB. This strategy enables the SSCB to dynamically respond to various fault conditions (e.g., low impedance faults, high impedance faults, fast fault recovery), significantly enhancing the reliability and performance of the EAP system.
- 2) A two-level fault detection strategy by combining fast pre-desaturation and desaturation detection is proposed for the SSCB. This improves fault detection speed and accuracy by addressing both fault current and its rate of change, enhancing the protection system reliability.
- 3) Four distinct operational modes are proposed for the SSCB—normal, precaution, i^2t , and instantaneous trip—allowing the system to dynamically adapt to various fault scenarios. In addition, the precaution mode ensures that the system current remains below the SCCR, offering enhanced protection for current-sensitive equipment.

The remainder of this article is organized as follows. Section II introduces an overview of the proposed SSCB. Section III presents two improved fault detection circuits suitable for the SSCB application. Section IV discusses the actuator of the proposed SSCB with four operation modes. Section V describes the coordinated controller design of the proposed SSCB. The implementation and verification of the proposed SSCB are discussed in Section VI. The proposed SSCB can seamlessly transfer among different operation modes based on various fault cases, and the experiment is conducted with an engineering prototype to validate the performance. Finally, Section VII concludes this article.

II. OVERVIEW OF THE PROPOSED SSCB

Fig. 2 overviews functional blocks of the proposed SSCB, including two fault detectors, three actuators, and a controller.

Two fault detectors are devised: PreDeSat-SSCB and DeSat-SSCB.

- 1) PreDeSat-SSCB relies on a di/dt -enabled fast fault current indicator, generating the precaution signal to alert SSCB of SC risk.
- 2) DeSat-SSCB depends on a modified, SSCB-suitable desaturation detection, generating the SC tripping signal to interrupt SSCB.

Once the fault is predicted or detected by the detectors, the controller identifies the fault type and determines the appropriate actions for the actuator based on the following operation modes.

- 1) *Normal Mode*: Under normal operation, SSCB's gate voltage stays at turn-on gate voltage (V_{GS-on}) through the actuator to conduct the normal system current with high efficiency.
- 2) *Precaution Mode*: Precaution mode is triggered by the precaution signal from PreDeSat-SSCB, and SSCB's gate voltage decreases to precaution voltage (V_{prec}) through the actuator to ensure the system's current (i_{sys}) cannot exceed the system's SCCR.
- 3) *i^2t Mode*: i^2t mode is triggered by an external i^2t detection unit, and SSCB's gate voltage decreases to turn-off voltage (V_{GS-off}) through the actuator to interrupt the low current fault. The interruption time depends on the power device's intrinsic thermal limit and the current i_{sys} .
- 4) *Instantaneous Trip Mode*: Instantaneous trip mode is triggered by the SC tripping signal from the DeSat-SSCB. SSCB's gate voltage decreases to V_{GS-off} through the actuator to interrupt the current at SSCB's SC tripping threshold.

By leveraging the different operation modes, the controller can enable SSCB to take different actions according to different fault cases, as summarized in Table III. Once di/dt -based fast fault current indicator (PreDeSat-SSCB) generates the precaution signal, SSCB enters precaution mode to ensure the maximum i_{sys} cannot exceed the system's SCCR and standby. Then, based on different fault cases, SSCB is controlled to enter different operation modes, as illustrated in Fig. 3.

III. FAULT DETECTOR

A fast and reliable fault detection method is crucial for the effective operation of SSCBs. Current sensors and desaturation detection (DeSat) [13], [28] are widely used to detect short-circuit events in SSCBs. Due to DeSat's fast response, simplicity, flexibility, and low cost, it is a popular choice

TABLE III
SSCB'S ACTIONS IN DIFFERENT FAULT CASES

Fault scenarios	Actions
Low impedance fault (System's fault current approaches SCCR and hits DeSat-SSCB's tripping threshold)	Step 1: Normal mode → Precaution mode Step 2: When i_{sys} hits SSCB tripping threshold, SSCB enters instantaneous trip mode to interrupt the fault current.
High impedance fault (System's fault current lower than SCCR)	Step 1: Normal mode → Precaution mode Step 2: Based on the power device's thermal performance, SSCB is interrupted by the i^2t signal from the external control unit.
Short-time overcurrent (System's start-up, fault fast recovery)	Step 1: Normal mode → Precaution mode Step 2: Back from precaution mode to normal mode, once the i_{sys} is back to normal.

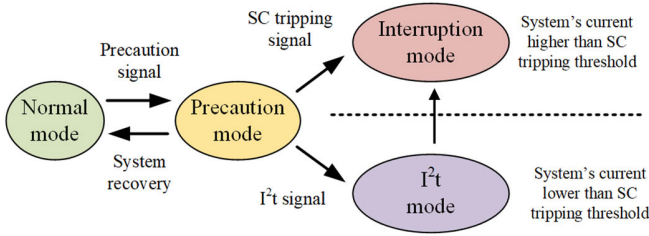


Fig. 3. Operation modes transformation for the proposed SSCB.

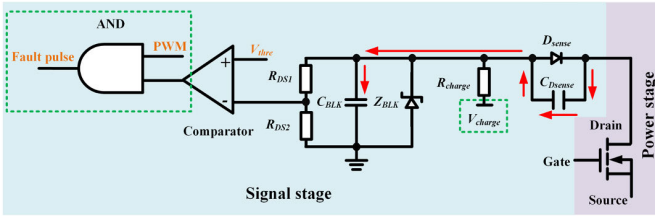


Fig. 4. Desaturation detection for SSCB.

for fault detection. DeSat is also commonly applied to the gate drive design of PWM-based converters to protect power semiconductors [29]. However, to better serve SSCBs, a modified DeSat detection circuit specifically tailored for SSCB applications is preferred.

A. Desaturation Detection for SSCB

The modified schematic of desaturation detection for SSCB (DeSat-SSCB) is displayed in Fig. 4. It mainly comprises drain–source voltage v_{DS} sensing diode D_{sense} , v_{DS} related resistor divider R_{DS1} and R_{DS2} , charging resistor R_{charge} , blanking capacitor C_{BLK} , voltage comparator, comparator protection diode Z_{BLK} . One of the most critical design considerations for desaturation detection is the selection of the D_{sense} . Besides the basic requirement that the breakdown voltage of D_{sense} should be higher than the maximum v_{DS} of the power device, the parasitic capacitance of the v_{DS} should be as small as possible. As shown in Fig. 4, the high dv_{DS}/dt can induce a charging current for C_{BLK} through C_{Dsense} , likely triggering the protection falsely. Thus, to mitigate the impact of the displacement current from D_{sense} , the capacitance of C_{BLK}

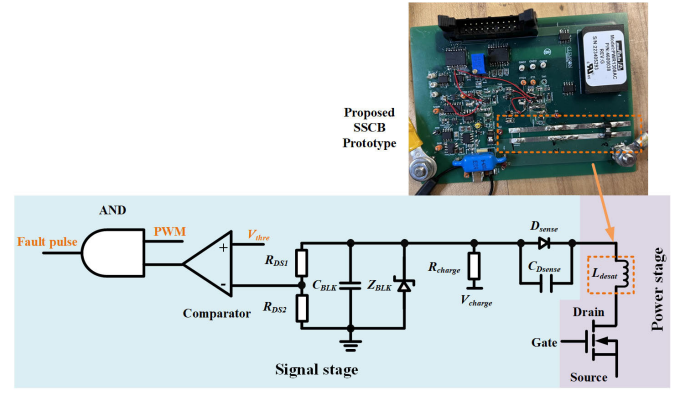


Fig. 5. Fast desaturation detection for SSCB.

should be significantly larger than the junction capacitance of D_{sense} . In this design, the GAP3SLT33-220FP, with a C_{Dsense} of 5 pF, is selected as the detection diode. To improve noise immunity, C_{BLK} is chosen to be 6800 pF, which is over 1000 times larger than C_{Dsense} .

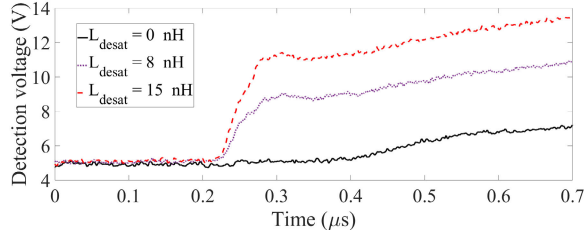
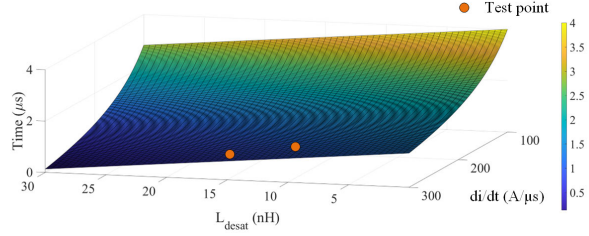
Different from traditional desaturation detection for a PWM-based converter (DeSat-PWM), a fixed voltage V_{charge} (e.g., 15 V) charges C_{BLK} . An additional “AND” gate connecting with the output of the comparator and PWM is required to avoid false overcurrent detection when the power device is off. The modified desaturation detection for SSCB mainly provides two benefits.

- 1) For DeSat-PWM, the voltage across the blanking capacitor v_{BLK} cannot exceed V_{GS-on} . For DeSat-SSCB, with a separate and more flexible charging source V_{charge} , there will be more margin for DeSat-SSCB to set the protection threshold, which is especially beneficial for GaN transistors with low V_{GS-on} (e.g., 6 V).
- 2) For the discrete components-based DeSat [30], once the fault event is detected, the gate drive IC (e.g., IXDD609) will be disabled, causing a high impedance at some gate drive IC output. Hence, for DeSat-PWM, the charging source of R_{charge} , which is determined by the gate drive output, becomes floated without steady potential. However, for DeSat-SSCB, with the fixed charging source V_{charge} for R_{charge} , its operation remains constant regardless of the gate drive IC.

In the proposed SSCB, the modified DeSat-SSCB is leveraged to detect if i_{sys} reaches SSCB's tripping threshold. The blanking time of the DeSat-SSCB is determined by the RC circuit charging time constant, which is the product of R_{charge} and C_{BLK} (i.e., $R_{charge} \times C_{BLK}$).

B. Fast Predesaturation Detection for SSCB

It is “ideal” to predict the fault event, allowing SSCB to be cautious about the modes' transition under different fault cases. With this, based on the DeSat-SSCB, a fast predesaturation detection for SSCB (PreDeSat-SSCB) is proposed in Fig. 5. Introducing an extra inductance L_{desat} in the power loop (e.g., power loop parasitics), the sensing voltage is also contributed by $L_{desat} \times di/dt$. Notably, L_{desat} is not a dedicated inductor design, instead the proposed SSCB simply leverages

Fig. 6. Waveform of detection voltage under different L_{desat} .Fig. 7. Relationship among L_{desat} , di/dt , and response time.

the parasitic inductance already existing in the hardware design through PCB trace. Because severe SC fault is usually associated with high di/dt , the proposed PreDeSat-SSCB is more sensitive to the SC fault not only due to the absolute fault current value but also its di/dt trend. With the L_{desat} , the detection voltage for PreDeSat-SSCB can be rewritten as

$$v_{\text{detect}} = v_{\text{DS}} + L_{\text{desat}} \frac{di_D}{dt} \quad (1)$$

where i_D is the current flowing through the power device's drain-source terminal.

Fig. 6 shows the test waveform of detection voltage with different L_{desat} (e.g., 0, 8, and 15 nH) under the di_D/dt of 300 A/μs. It can be observed that the PreDeSat-SSCB can detect the influence of di/dt . Fig. 7 illustrates the relationship among L_{desat} , di/dt , and the time of the detection voltage (using 10 V as an example threshold). The relationship shows that the larger the L_{desat} and the faster di/dt , the shorter the detection response time. Under the condition of $di/dt = 300$ A/μs, compared with $L_{\text{desat}} = 0$ nH, $L_{\text{desat}} = 15$ nH can speed up the response speed by 10×.

In the proposed SSCB, the PreDeSat-SSCB is leveraged to detect whether the SSCB is under the SC risk. It generates the fault signal once the system has a relatively high current and di/dt , which can help SSCB “predict” the fault and be alerted. Once the protection signal associated with the PreDeSat-SSCB is tripped, SSCB enters the precaution mode. It is noted that the proposed PreDeSat-SSCB cannot be leveraged to the power converter; otherwise, it may trigger the protection during the switching transition under its normal PWM operation.

Fig. 8 demonstrates the protection threshold settings associated with PreDeSat-SSCB and DeSat-SSCB, respectively, where the blue curve is the V – I curve with the gate voltage V_{prec} and the black curve is the V – I curve with the gate voltage $V_{\text{GS-on}}$. To allow the PreDeSat-SSCB to identify the potential SC fault promptly, its threshold voltage V_{thr1} is relatively small,

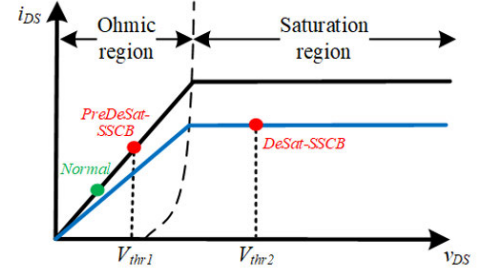


Fig. 8. Detection points for PreDeSat-SSCB and DeSat-SSCB.

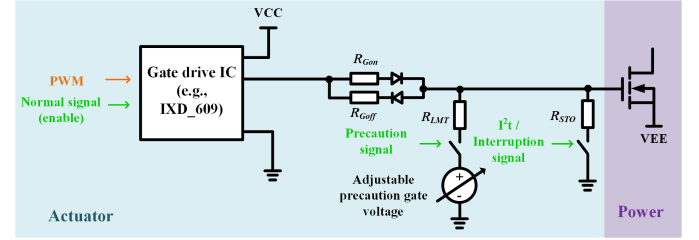


Fig. 9. Proposed actuator and its circuit implementation.

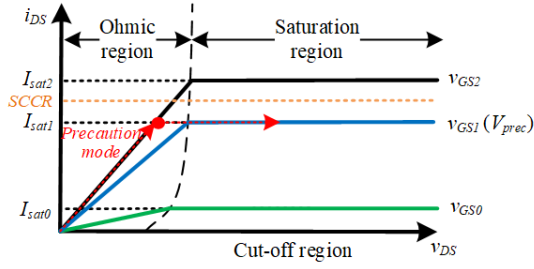
and the corresponding device current is much higher than the normal operating current but lower than the saturation current. For DeSat-SSCB, a relatively high threshold voltage V_{thr2} is selected to ensure SSCB's interruption when the device is fully saturated. Compared to traditional desaturation detection, the combination of PreDeSat-SSCB and DeSat-SSCB in our two-level detection system enhances response speed and reduces the likelihood of false triggering of the protection.

C. i^2t Detection

The essential functions of SSCBs include SC and overload protection [31]. These two protections are triggered based on different fault current levels. The instantaneous trip region is designated for SC protection, where the protection mechanism triggers immediately when the detected current exceeds the SC current threshold. The overload trip, or i^2t trip region, is designed for overcurrent protection. The i^2t protection activates after the overcurrent condition persists for a specified duration. By leveraging the thermal performance of the SSCB's power device, the external i^2t detection unit predefines an i^2t curve for SSCB. Once SSCB standing time for a low fault current achieves the i^2t preset time, an i^2t signal will be sent to SSCB to interrupt SSCB. Notably, the design of the i^2t protection curve is based on the article [31]. The proposed SSCB received a signal from the external i^2t detection unit to trigger the i^2t mode.

IV. ACTUATOR

As mentioned in Section II, the proposed SSCB has four operation modes: normal mode, precaution mode, i^2t mode, and instantaneous trip mode. According to different operation modes, SSCB's actuator circuitries in Fig. 9 control gate voltage differently to manage the power device characteristics of SSCB.

Fig. 10. Typical V - I curve for a power device.

A. Normal Mode

When there is no fault, SSCB is kept in normal mode. The normal mode signal controls the enable pin of the gate drive IC. Once the normal mode signal is effective, the gate drive IC will be enabled, and the power device's gate voltage will be under the control of the PWM signal. Once the normal mode signal is ineffective, the gate drive IC will be disabled as a high impedance, and the gate voltage will be dominated by the other operation mode.

B. Precaution Mode

Once the potential SC fault is detected by the PreDeSat-SSCB-based sensitive detector, SSCB enters precaution mode. Precaution mode leverages the V - I curve characteristic to avoid i_{sys} being higher than SSCR. As shown in Fig. 10, once SSCB enters precaution mode, the power device's gate voltage decreases from V_{GS-on} to V_{prec} , where blue curve is the V - I curve with the gate voltage V_{prec} and black curve is the V - I curve with the gate voltage V_{GS-on} . If the SC fault current approaches SSCR, the power device would be forced to enter the saturation region, and the i_{sys} is limited to the saturation current predefined by lower gate voltage V_{prec} . In other words, the V_{prec} should be selected in such a way that the corresponding saturation current is lower than SSCR. It is noted that without the current limitation function, system fault current may exceed SSCR and endanger system operation. Meanwhile, because the saturation current is a function of the gate voltage, by adjusting V_{prec} , the current limitation level can also be conveniently adjusted to satisfy the system's SSCR requirement. Moreover, for the surge current during the system start-up, we can also ensure the system current remains below SSCR by setting SSCB at precaution mode during system initialization. The current limitation level can be estimated by

$$I_{sat} = g_m(v_{gs} - v_{th}) \quad (2)$$

where g_m is the transconductance of the power semiconductor and v_{th} is the turn-on threshold voltage.

Programmable V_{prec} can be achieved by the circuit in Fig. 11. It consists of one digital potentiometer (DPM) and a low dropout (LDO) voltage regulator with adjustable output. The wiper position can change the resistance value of the DPM. The wiper is located between the point of A (POA) and the point of B (POB). The input of the DPM determines the address of the wiper. For the DPM, to control the location of the wiper, the upper-level controller sends the address of

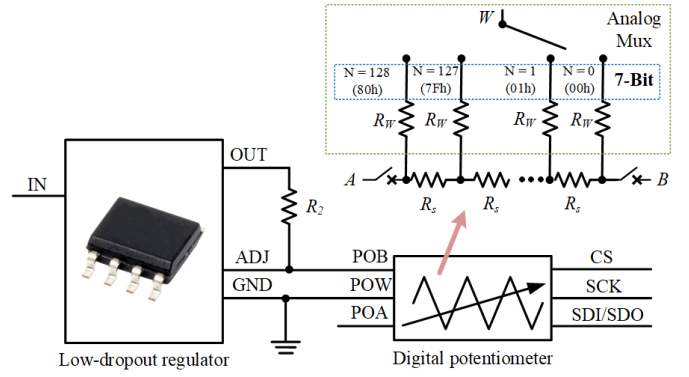


Fig. 11. Schematic of the adjustable precaution voltage circuit.

the wiper to the DPM through serial peripheral interface (SPI) communication. SPI communication relies on the chip selection (CS) signal, the clock (SCK) signal, and the serial data in/out (SDI/SDO) signal to complete the data transmission. With the SPI control, the resistance between POB and the point of the wiper (POW) can be expressed as

$$R_{WB} = \frac{R_{AB} \times N}{N_{step}} \quad (3)$$

where N_{step} is the total step for a DPM (e.g., a 7-bit DPM has 127 steps); N is the wiper address (0–127 for a 7-bit DPM) sent by the SPI communication; R_{AB} is the resistance between POA and POB.

An adjustable LDO to generate the required gate voltage is also important to realize the online adjustable gate voltage control. The output of the adjustable LDO can be derived as follows:

$$V_{LDO_out} = V_{adj} \times \left(1 + \frac{R_{WB}}{R_2}\right) \quad (4)$$

where V_{adj} (1.2–1.3 V for most adjustable LDOs) is the voltage across the ADJ and GND of the LDO, providing the reference voltage for the output; R_2 is the resistor between the OUT and ADJ pin of the LDO.

Based on the output equation of the adjustable LDO, with the fixed R_2 , the output of the LDO can be controlled by the DPM through SPI communication.

C. i^2t Mode

Once SSCB receives an i^2t detection enable signal from the external control unit, SSCB enters i^2t mode. During the operation of SSCB, the external i^2t detection and protection unit monitors i_{sys} and calculates the i^2t value [27]. Based on the predefined device's i^2t capability curve, once the sensed i^2t exceeds the devices' capability, an i^2t fault is tripped, and its protection signal is transmitted to SSCB. Once entering i^2t mode, SSCB's gate voltage can decrease from V_{prec} to V_{GS-off} to avoid damage to the power device by a thermal issue.

D. Instantaneous Trip Mode

When i_{sys} hits SSCB tripping threshold, the proposed DeSat-SSCB generates the interruption signal, and SSCB

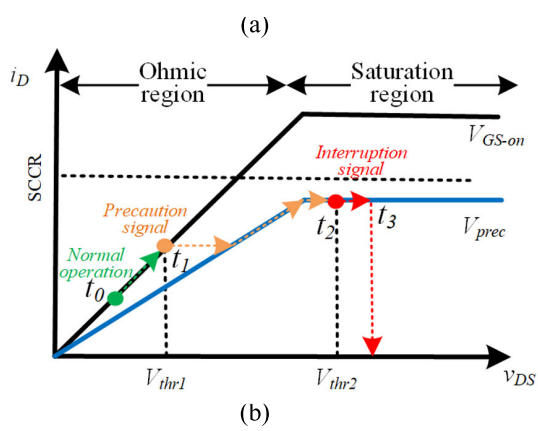
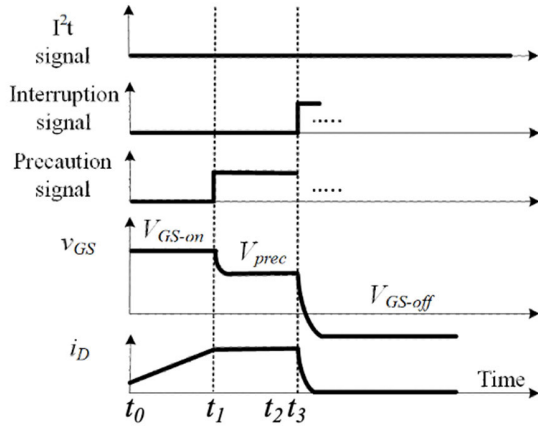


Fig. 12. Operation principle under the low impedance fault. (a) Control logic. (b) Operating trajectory.

TABLE IV
DETAIL OF DIFFERENT OPERATION MODES

Mode	Detector			Actuator		
	DeSat	PreDeSat	Ext. i^2t	V_{GS-on}	V_{prec}	V_{GS-off}
Normal	×	×	×	○	×	×
Precaution	×	○	×	×	○	×
I^2t	×	×	○	×	×	○
Inst. trip	○	×	×	×	×	○

○: with ×: without

enters the instantaneous trip mode. Once entering the instantaneous trip mode, SSCB's gate voltage can decrease from V_{prec} to V_{GS-off} to turn off power devices.

Table IV summarizes the details of different operation modes.

V. COORDINATED CONTROL OF SYSTEM-FRIENDLY SSCB

As mentioned in Table III, a coordinated controller has been developed to help SSCB take various actions under different fault cases. As a reminder, the threshold V_{thr2} of the DeSat-SSCB corresponds to the SSCB's tripping current threshold, and V_{prec} makes sure that once the SSCB enters the precaution mode, the system current will be limited to lower than SCCR.

A. Low Impedance Fault

Fig. 12 illustrates the control logic and device $V-I$ trajectory under the low impedance fault. The proposed

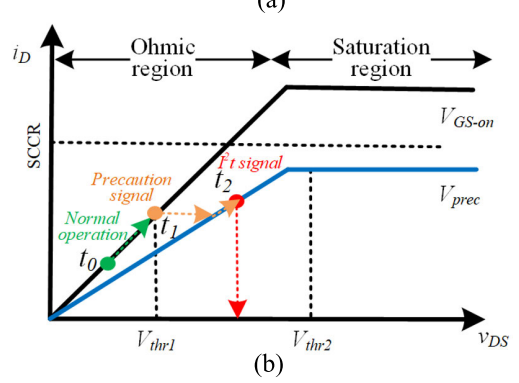
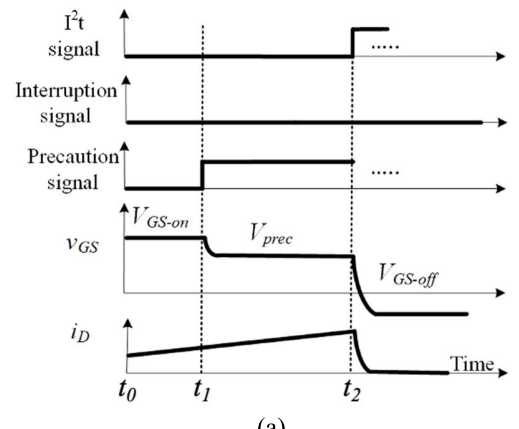


Fig. 13. Operation principle under the high impedance fault. (a) Control logic. (b) Operating trajectory.

PreDeSat-SSCB and DeSat-SSCB generate the precaution signal and interruption signal, respectively. PreDeSat-SSCB detects the SC event faster. Once PreDeSat-SSCB generates an effective precaution signal at t_1 , v_{GS} will drop from V_{GS-on} to V_{prec} to ensure i_{sys} is lower than SCCR. Then, at t_2 , with the system fault current approaches SCCR, DeSat-SSCB is triggered to generate an effective interruption signal, and SSCB transitions from the precaution mode to the instantaneous trip mode. With hundreds of nanosecond blanking time, SSCB interrupts the system fault current at t_3 . Due to the sensitive PreDeSat-SSCB detection, SSCB transitions to the precaution mode promptly, allowing v_{GS} to decrease to V_{prec} in time to limit the system fault current below SCCR.

B. High Impedance Fault

Fig. 13 illustrates the control logic and device $V-I$ trajectory under the high impedance fault. Similar to low impedance fault, once PreDeSat-SSCB generates an effective precaution signal at t_1 , v_{GS} drops from V_{GS-on} to V_{prec} . Because of the high impedance fault, it is assumed that the system fault current is lower than SCCR; therefore, DeSat-SSCB fault detection will not be enabled. In this case, the external i^2t control unit kicks in to let SSCB enter i^2t mode from precaution mode and interrupts the fault at t_2 . This operation scenario allows SSCB to stand by with a relatively low fault current, offering time for system-level protection coordination, which is particularly beneficial to the current sensitive EAP system with high penetration of power electronics [32].

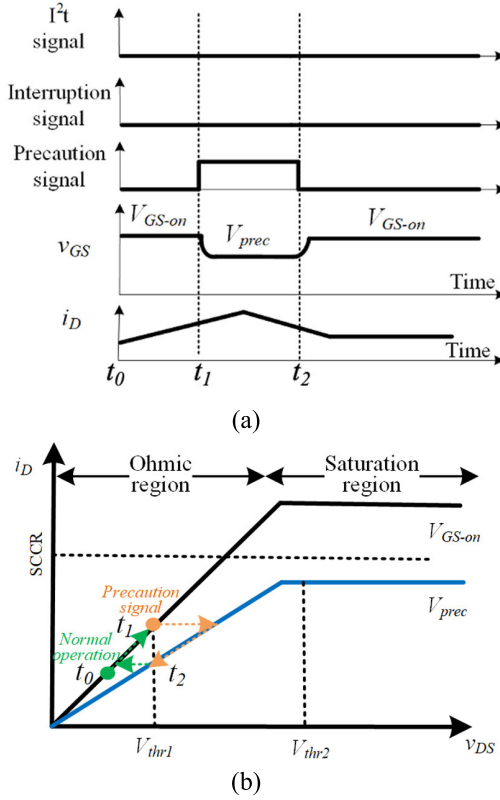


Fig. 14. Operation principle under the fault recovery. (a) Control logic. (b) Operating trajectory.

C. Fault Fast Recovery Case

The control logic and SSCB trajectory under the fault fast recovery case are shown in Fig. 14. Assume PreDeSat-SSCB generates an effective precaution signal at t_1 due to a fault event, lowering v_{GS} to be V_{prec} . However, the fault is recovered between t_1 and t_2 (either the fault disappears automatically or the fault is cleared by the local SSCB), and the system's current is back to the normal value at t_2 . Because the fault time is short enough not to trigger the i^2t mode, the precaution signal would disappear, and v_{GS} is back to normal V_{GS-on} . This operation scenario enables SSCB to have fault ride-through capability. The short-time overcurrent fault or event cannot influence the operation of SSCB.

VI. SYSTEM-FRIENDLY SSCB IMPLEMENTATION AND VERIFICATION

Integration with Sections III–V, Fig. 15 shows the schematic of the proposed SSCB.

A. Implementation

1) *Detector*: The detection circuit is composed of DeSat-SSCB and PreDeSat-SSCB. The sensitive PreDeSat-SSCB is leveraged to detect the potential SC fault. Meanwhile, the DeSat-SSCB is leveraged to detect SSCB's SC tripping threshold current. To make the precaution mode more sensitive to potential SC fault, the threshold voltage of PreDeSat-SSCB V_{thr1} should be relatively small. Meanwhile, the threshold voltage of DeSat-SSCB V_{thr2} needs to fit with SSCB's SC tripping threshold current.

2) *Controller*: The controller comprises several logic gate ICs to generate SSCB's control signal. The precaution pulse is generated by PreDeSat-SSCB, while DeSat-SSCB generates the interruption pulse. The i^2t signal is from the external control unit. The generated precaution pulse is sent to an AND gate (AND 3). Meanwhile, the generated interruption pulse is sent to an OR gate (OR 1) with the external i^2t signal. The output of OR 1 will be sent to a latch-up circuit to generate the turn-off signal to interrupt the system's current. The precaution signal will be AND with the inversed turn-off signal generating the precaution mode control signal, to avoid the conflict between limitation and turn-off signal. Finally, the precaution mode control signal and turn-off signal will be OR together to generate the enable signal for the gate drive IC (normal mode control signal). The enable signal is leveraged to control the gate drive IC for the power device.

3) *Actuator*: The actuator circuit mainly comprises the gate drive IC, precaution voltage control branch, and turn-off control branch. The gate drive IC is leveraged to control the power device under normal conditions. Once SSCB enters precaution mode, the gate drive IC will be disabled. The precaution signal can conduct the precaution voltage control branch, and v_{GS} can drop to V_{prec} to ensure i_{sys} cannot be higher than SCCR. Meanwhile, the turn-off signal generated from the interruption signal and i^2t signal can conduct the turn-off control branch, and v_{GS} can drop to turn-off voltage to interrupt the system's fault current.

B. Experimental Verification

The actual power rating for SSCBs in EAP systems can reach up to 2 kV/1 kA [33]. In most cases, the design can be modular. Increasing power capacity can be achieved by connecting power semiconductors in parallel and series. Yang et al. [34] and Dam et al. [35] have studied the design of high power SSCB with modular architecture. Therefore, as shown in Fig. 16, for the purposes of this demonstration, a single module with a downscaled rating GaN-based 200-V/10-A SSCB was adopted to validate the concept. Table V summarizes the key parameters.

Compared to SiC-based power semiconductors, GaN offers a lower $R_{DS(on)}$, which reduces conduction losses and increases efficiency. In addition, GaN devices are grown on standard, widely available silicon substrates, providing a significant cost advantage over SiC, which relies on more specialized and expensive materials. Furthermore, in aircraft cooling systems, cryogenic temperatures can help reduce energy consumption and enhance overall system efficiency. GaN also performs exceptionally well in cryogenic environments, making it a promising technology in the emerging trend of aviation applications [36].

The snubber circuit design follows the approach outlined in [37]. The primary purpose of the snubber circuit is to reduce the dv/dt and minimize turn-off energy dissipation. Most of the components, such as digital isolators and isolated power supplies, can withstand at least 50 V/ns. During turn-off, the fault current transfers from the GaN device to the snubber

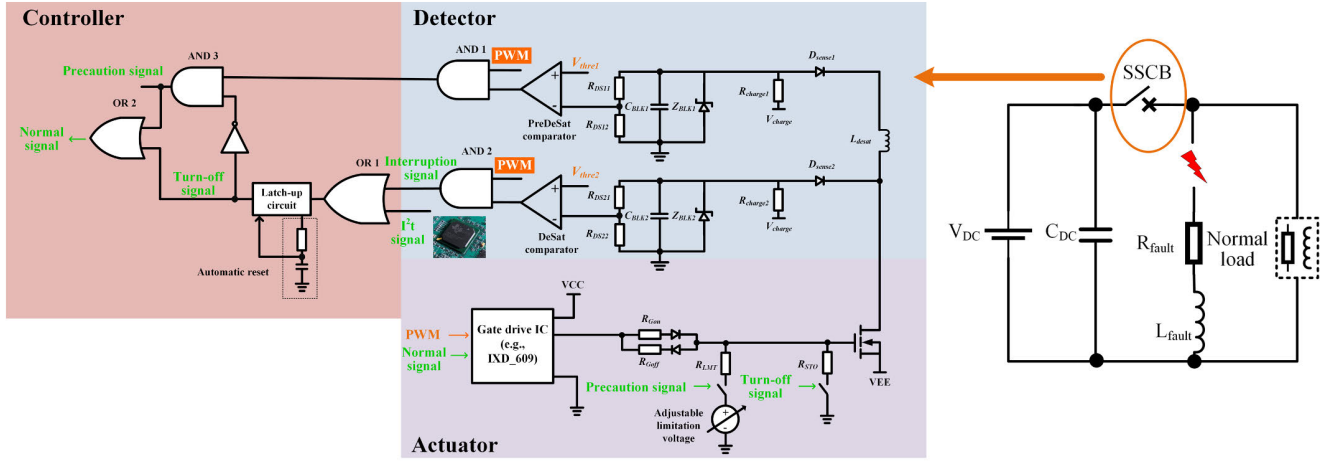


Fig. 15. Schematic of the proposed SSCB.

TABLE V
PARAMETERS OF THE SSCB PROTOTYPE

Parameters	Value
Power device	GS66516T
Snubber	150nF, 0.5Ω
TVS	AK3-430
SCCR	150A
DC bus voltage	200V
Turn-on voltage	6V
Turn-off voltage	-3V
Precaution voltage	4V
Gate Drive IC	IXDD609
D _{sense}	GAP3SLT33
C _{BLK}	6800pF
Comparator	TLV3201AIDBVT
LDO for adjustable gate voltage	ADP7104ARDZ-R7
Digital potentiometer	MCP4131
R _{DS11} , R _{DS12} , R _{DS21} , R _{DS22}	10k (need to be modified for different case)
C _{BLK1} , C _{BLK2}	6800 pF
R _{charge1} , R _{charge2}	50ohm
R _{STO}	5ohm
R _{Gon}	2ohm
R _{Goff}	2ohm
R _{LMT}	5ohm

branch, causing the voltage across the snubber capacitor to increase as it charges. The dv/dt can be controlled by selecting an appropriate snubber capacitor. However, due to circuit parasitics, the actual dv/dt during switching tends to be higher than expected due to parasitic oscillations, necessitating the use of a larger capacitance. To account for these effects, a SPICE simulation was performed on the designed breaker, revealing that a capacitance of 150 nF is required to keep dv/dt below 50 V/ns.

The test platform is mainly composed of the dc power supply V_{DC} , fault trigger breaker, system fault impedance Z_{fault} , and the proposed SSCB. The normal load is assumed to be infinite. A separate fault trigger breaker is introduced to create the SC fault events. Once the fault trigger breaker is conducted, it can be assumed that the SC occurs. Z_{fault} can emulate the fault system impedance once the SC occurs. The response of the proposed SSCB to different SC fault cases could be tested by adjusting Z_{fault} .

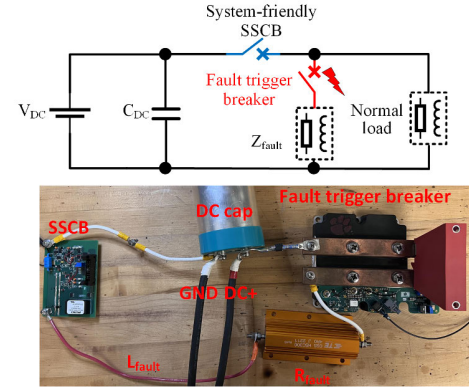
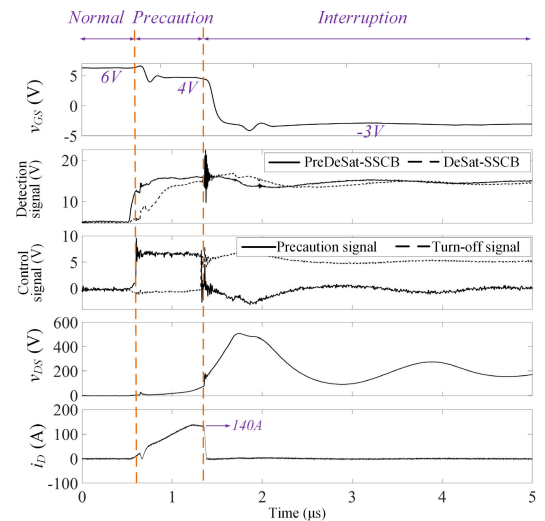


Fig. 16. Test platform for the proposed SSCB.

Fig. 17. Test waveform under the low impedance fault ($L_{fault} = 0.7 \mu\text{H}$).

1) *Low Impedance Fault*: Fig. 17 illustrates the waveform of the SSCB when the fault system inductance L_{fault} is $0.7 \mu\text{H}$, which is primarily contributed by interconnecting wire instead of a dedicated inductor. The SC fault happens at

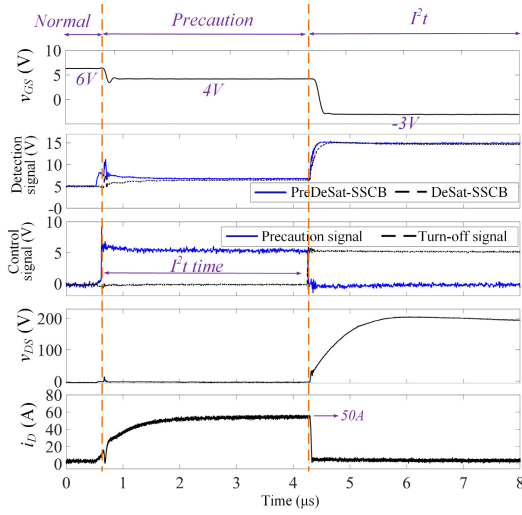


Fig. 18. Test waveform under the high impedance fault ($L_{\text{fault}} = 2 \mu\text{H}$, $R_{\text{fault}} = 4 \Omega$).

about $0.5 \mu\text{s}$, and v_{GS} is 6 V. Then, at approximately $0.5 \mu\text{s}$, the PreDeSat-SSCB detects the potential SC, SSCB enters precaution mode, and v_{GS} drops to 4 V, which corresponds to the predefined SCCR in Table V. DeSat-SSCB detects the overcurrent at about $1.4 \mu\text{s}$, SSCB enters instantaneous trip mode, and v_{GS} drops from 4 to -3 V to interrupt the system's current. Based on the waveform, it can be found that the proposed SSCB can interrupt a low impedance fault in $1 \mu\text{s}$; meanwhile, the current flowing through the power device i_D is limited by the precaution voltage to 140 A stably.

2) *High Impedance Fault*: Fig. 18 illustrates the waveform of the SSCB with $2\text{-}\mu\text{H}$ system inductance L_{fault} and 4Ω system resistance R_{fault} . The system's maximum fault current is limited to 50 A by R_{fault} . The SC fault happens at $0.5 \mu\text{s}$, and v_{GS} is 6 V. At $0.8 \mu\text{s}$, the PreDeSat-SSCB detects the potential SC, SSCB enters precaution mode, and v_{GS} drops to 4 V. Approximately $4 \mu\text{s}$ later, SSCB receives an emulated i^2t signal for the sake of test and enters i^2t mode. i^2t signal generates a turn-off signal, and v_{GS} drops from 4 to -3 V to interrupt the system's fault current. Because the v_{DS} does not achieve the threshold voltage for DeSat-SSCB, DeSat-SSCB is not triggered in this case. The test results agree with the analysis above.

3) *Fault Fast Recovery Case*: Fig. 19 illustrates the waveform of the SSCB under a fault fast recovery case. The SC fault happens at $0.5 \mu\text{s}$ with 6-V v_{GS} . Then, at $1 \mu\text{s}$, the PreDeSat-SSCB detects the potential SC, and SSCB enters precaution mode with 4-V v_{GS} . Then, the fault recovers at $2.2 \mu\text{s}$. With the fault recovery, the system's current decreases. Accordingly, PreDeSat-SSCB detected v_{DS} decreases and becomes lower than PreDeSat-SSCB's threshold voltage. Then, the precaution signal disappears, and v_{GS} increases from 4 to 6 V to resume the system's normal operation. During this process, because the fault time is shorter than i^2t time and the system's fault current is not high enough, i^2t and instantaneous trip modes are not triggered.

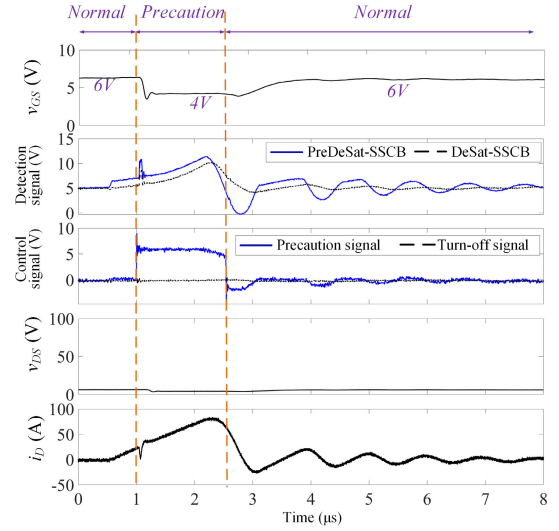


Fig. 19. Test waveform under the fault fast recovery case.

C. Signal Delay Consideration

The typical signal delay time for the designed detection circuit is approximately 40 ns, including gate logic stages and one comparator. For DeSat-SSCB, with a system inductance of $0.7 \mu\text{H}$, representing a very low impedance fault with high di/dt , the response time is around $1 \mu\text{s}$. Therefore, the 40-ns signal delay can be considered negligible in the context of DeSat-SSCB. For PreDeSat-SSCB, because of the existence of L_{desat} , the response time can be about 50–100 ns. However, once the PreDeSat-SSCB is triggered, SSCB will enter the precaution mode to limit the system current. For the case in Fig. 17, it needs to take about $1 \mu\text{s}$ to achieve the SCCR (140 A). Thus, the 40-ns signal delay will not influence the operation of the precaution mode to limit the system current. Accordingly, the signal delay will not influence the operation of the proposed strategy.

D. DeSat Discussion for Modular SSCB

For devices connected in parallel, each device is equipped with its own DeSat protection, which directly turns off the device when a fault is detected. In this scenario, if some parallel devices have not yet tripped their DeSat, they will momentarily share the additional current from the device that has already been turned off. This increased current will trigger their DeSat protection shortly thereafter. As long as the DeSat response is fast enough, all parallel devices will be adequately protected. In series configurations, all devices inherently share the same current, meaning only one device with a DeSat circuit is sufficient to detect the system fault current. Therefore, implementing DeSat protection does not introduce additional complexity in modular designs.

VII. CONCLUSION

An intelligent dc SSCB with integrated fault management capabilities is developed and demonstrated. The key takeaways and findings are summarized below.

- 1) Two fault current detection circuits, DeSat-SSCB and PreDesat-SSCB, are suitable for the SSCB application. Particularly, the PreDesat-SSCB, considering both the fault current absolute value and di/dt , results in the increased fault detection response speed up by $10\times$ as compared to conventional desaturation detection based on the test.
- 2) Four operation modes, including normal mode, precaution mode, i^2t mode, and instantaneous trip mode, are offered: a) normal mode ensures the efficient operation of the SSCB; b) precaution mode ensures the system's fault current lower than SCCR; c) i^2t mode helps SSCB interrupt low fault current; and d) instantaneous trip mode allows SSCB's interruption at SCCR in 100 s of ns to $1-\mu$ s response time.
- 3) Coordinated control enables SSCB to take corresponding actions to various fault cases: a) once the SC and/or high di/dt event is detected through PreDesat-SSCB, the proposed SSCB lowers the power device gate voltage to ensure the system's maximum fault current below SCCR; b) under the low-impedance fault when the fault current approaches SCCR, the proposed SSCB based on DeSat-SSCB triggering signal can interrupt the fault within 200 ns to $1\ \mu$ s; c) under the high-impedance fault when the fault current does not exceed SCCR, the proposed SSCB can interrupt the fault based on an external i^2t detection and protection unit; and d) finally, the SSCB is capable of fast fault recovery, i.e., when the fault is cleared by a local SSCB and the protection signal by PreDesat-SSCB disappears, the SSCB is able to be back under the normal operation.

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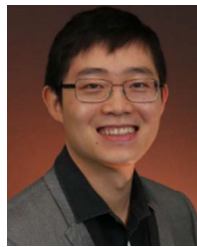


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